

Zaw Wana

Platform / Site Reliability Engineer — Kubernetes | Linux | Observability

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PROFESSIONAL SUMMARY

Certified Kubernetes Administrator running production container platforms and Linux fleets in enterprise and public-sector environments. Day-2 operations across the full cluster lifecycle: node lifecycle and etcd health, RBAC and SCC policy, network policies, routes and HAProxy ingress, resource quotas, and persistent storage. First-responder experience tracing failures from node to pod to service, with root-cause analysis fed back into runbooks, and ELK/Prometheus observability to catch failures before users do. Also owns the Linux OS layer beneath the platform — patching cycles, kernel and resource tuning, hardening, and storage — plus highly available data tiers and secrets through Vault and CyberArk. Currently supporting a government AI infrastructure programme under IM8-aligned security controls. Computing Science honours graduate, Singapore Institute of Technology.

CORE TECHNICAL SKILLS

Kubernetes & Containers	Kubernetes (CKA) — node lifecycle, etcd health, RBAC, network policies, resource quotas, PV/PVC & storage classes; OpenShift Container Platform (OCP 4.x) projects, Routes, SCCs, HAProxy ingress routing and timeout tuning, private registry & image mirroring; K3s, Helm, Docker
Reliability & Observability	Incident response and root-cause analysis, runbooks, ELK / Elasticsearch / Logstash / Kibana, OpenSearch, Prometheus, Grafana, Zabbix, log aggregation, alerting, capacity & performance monitoring
High Availability	MariaDB Galera (multi-master HA), HAProxy proxy tier for connection routing and health checking, backup & recovery, JVM/Java workload tuning (heap, Metaspace, GC flags, container memory budgeting)
Linux Platform	RHEL-family (RHEL/CentOS/Rocky) & Ubuntu administration, OS patching & kernel upgrades, reboot sequencing, dnf/yum package management, systemd, SELinux, firewalld, auditd, ulimits/cgroups, NFS & iSCSI/SMB storage mounts, LVM, Windows Server
Automation & IaC	Terraform, Ansible, Jenkins, GitLab CI, ArgoCD / GitOps, Git, Python, Bash, operational runbooks
Security & Hardening	CIS benchmark hardening, IM8-aligned controls, vulnerability remediation & CVE triage, SSH cipher/MAC hardening, HashiCorp Vault, CyberArk PAM, LDAP, Azure AD (Entra ID) IAM & RBAC, IAM, secrets management, IPsec VPN, DevSecOps
Data & Cloud	MySQL, MongoDB, Redis; AWS (EC2, S3, IAM, CloudWatch, VPC), Azure (Entra ID, RBAC, Policy), hybrid & on-prem integration, Proxmox virtualisation

WORK EXPERIENCE

DevSecOps / Platform Engineer — NCS

Nov 2025 — Present

Video intelligence and enterprise AI infrastructure platforms — on-prem estate, government security baseline.

- Operate the OpenShift Container Platform estate the workloads run on — projects and namespaces, RBAC and SCC policy, network policies and resource quotas, Routes and HAProxy ingress, and persistent storage via PV/PVC and storage classes
- Built and operate MariaDB on a Galera cluster — synchronous multi-master replication so the database survives node loss without failover downtime or manual promotion; front-ended by a proxy tier (HAProxy) for connection routing and health checking
- Operate ELK-based monitoring across cluster, systemd service, and MariaDB logs — aggregated into dashboards and alerting so failures surface before users report them
- Own the Linux OS layer beneath the platform — patching cycles, kernel and package management, resource limits, and storage mounts — sequenced so production services stay up through the change window
- Day-1 build and Day-2 operations on an enterprise AI infrastructure platform, plus Database Engineer ownership of schema design, tuning, and backup and recovery

- Provision platform infrastructure with Terraform so environments are described in code and rebuilt from it rather than reassembled by hand
- Apply and validate OS hardening against CIS-style benchmarks and government (IM8) security requirements; remediate vulnerability-scan findings and produce the supporting remediation evidence
- Manage privileged access with CyberArk and centralise service credentials in HashiCorp Vault, removing long-lived secrets from configuration files and node images
- Administer identity and access governance on Azure AD (Entra ID) — user/group lifecycle, RBAC role assignments, and Azure Policy enforcement — alongside on-prem LDAP
- Run LDAP-backed authentication and group authorisation across a mixed RHEL-family Linux and Windows Server estate
- Configure IPsec VPN tunnels so field mobile devices reach internal systems without exposing them publicly

DevOps / Infrastructure Engineer — Singtel

Sep 2024 — Jul 2025

Infrastructure, delivery, and operations behind Singtel's NaaS, 5G, and multi-access edge computing (MEC) products.

- Ran day-to-day operations across Linux servers and Kubernetes clusters as first responder — tracing failures from node to pod to service, with root-cause analysis feeding back into runbooks
- Administered Kubernetes clusters supporting edge workloads: node management, workload placement, resource limits, and standardised deployments packaged with Helm
- Implemented full-stack Zabbix monitoring across infrastructure, network, and application metrics
- Maintained and extended Jenkins CI/CD pipelines, and delivered application rollouts through ArgoCD — Git as source of truth, with cluster drift reconciled automatically instead of patched in place
- Troubleshoot and performance-tuned OpenSearch, MySQL, Redis, MongoDB, and container workloads
- Owned OS patching and update cycles across the Linux server fleet — keeping systems current and vulnerabilities closed without taking production services down
- Automated infrastructure provisioning with Terraform and daily Docker/virtualised operations with Ansible
- Wrote Python automation for security (IoC IP filtering) and operations (system health checks)

Software Engineer — Urban Farming

Oct 2022 — Apr 2023

IoT aggregation hub turning raw sensor metrics into crop-yield decisions. (Part-time, alongside degree)

- Structured the backend as Spring Boot microservices — ingestion, processing, and API as separate deployable services rather than one process — with a React/Node aggregation layer over real-time IoT sensor metrics
- Gave the farming team one source of truth to optimise irrigation, lighting, and climate control

PLATFORM ENGINEERING — SELF-DIRECTED

- Production-grade K3s platform (homelab, continuously operated): multi-node cluster run through real upgrade, certificate-rotation, and failure scenarios, with GitOps delivery, ingress and TLS termination, persistent storage classes, RBAC and namespace quotas, private image registry, and a full Prometheus/Grafana + ELK observability stack — built to enterprise patterns
- HA data tier: MariaDB Galera cluster behind an evaluated proxy layer (HAProxy / MaxScale / ProxySQL), with backup, restore, and node-loss drills
- Bash-based CKA exam simulator running against a disposable k3d sandbox; AWS serverless projects for SAA preparation

CERTIFICATIONS & EDUCATION

Certified Kubernetes Administrator (CKA) — Cloud Native Computing Foundation, Aug 2026 · ID LF-ijwzjz1ojo

BSc Computing Science (Honours) — Singapore Institute of Technology

Aug 2022 — Apr 2025

Diploma in Business Process with Systems Engineering — Temasek Polytechnic

Apr 2017 — Feb 2020